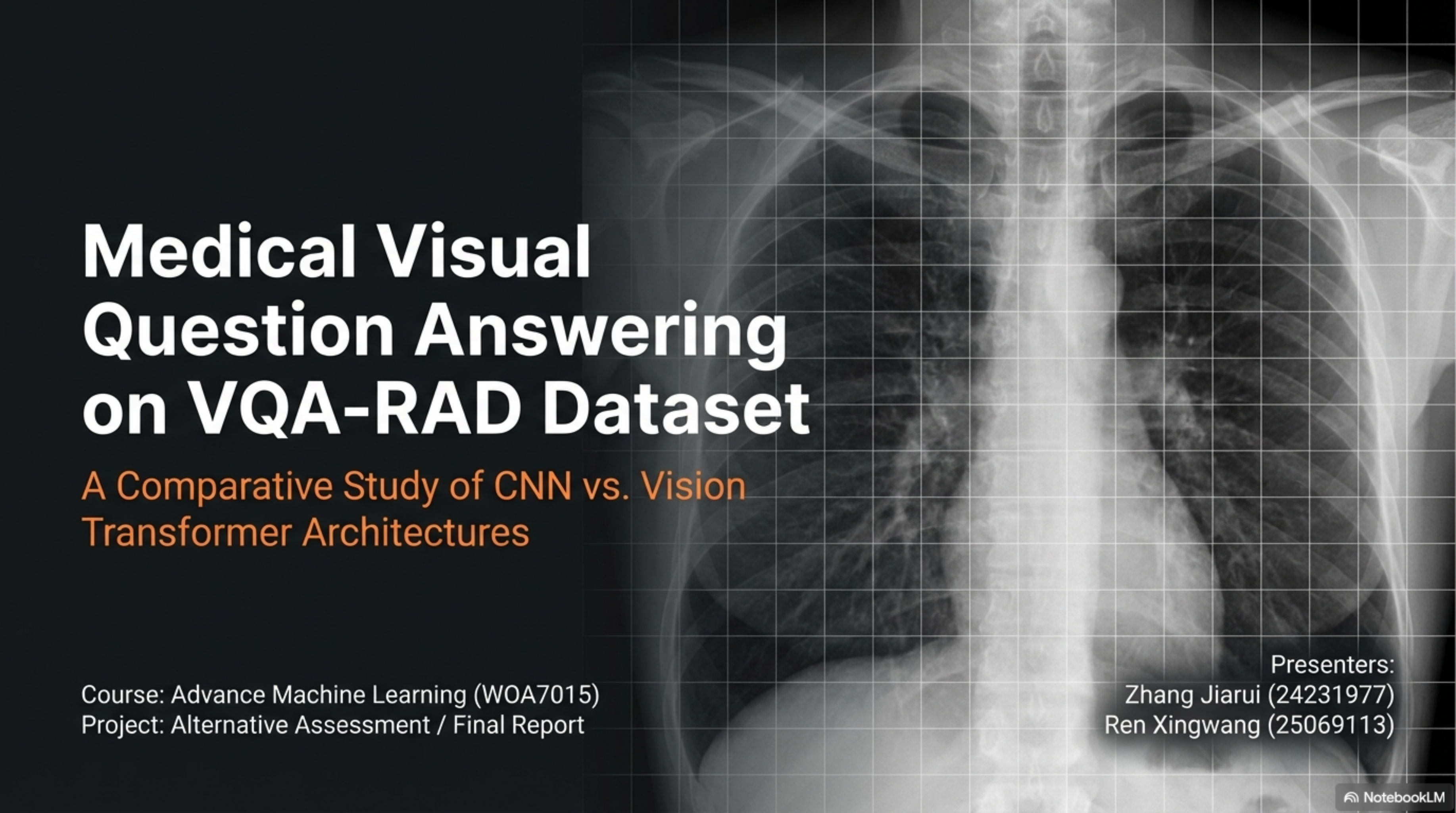




Medical Visual Question Answering on VQA-RAD Dataset

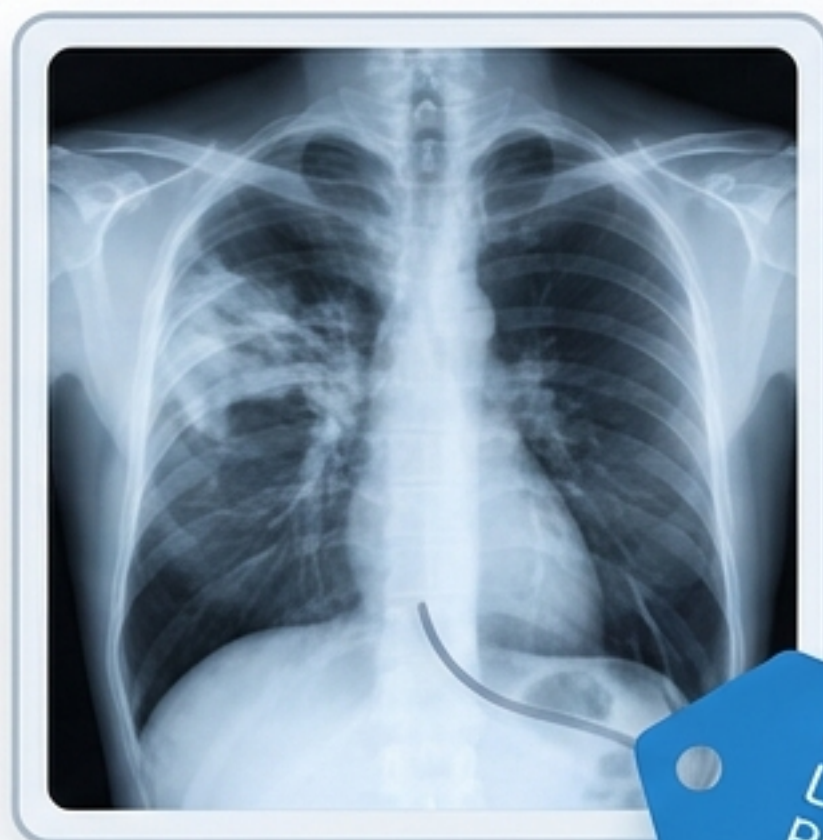
A Comparative Study of CNN vs. Vision
Transformer Architectures

Course: Advance Machine Learning (WOA7015)
Project: Alternative Assessment / Final Report

Presenters:
Zhang Jiarui (24231977)
Ren Xingwang (25069113)

Moving Beyond Simple Classification

THE OLD WAY



Traditional AI:
Static Classification

Scrub Blue
Label:
Pneumonia
(90%)

The Goal:
From Labeling to
Understanding.

THE NEW WAY



Med-VQA:
Reasoning + Interaction

Research Question: Can the global context of Transformers outperform local CNN features in radiology?

The Challenge: VQA-RAD Dataset Constraints

Key Stats

216

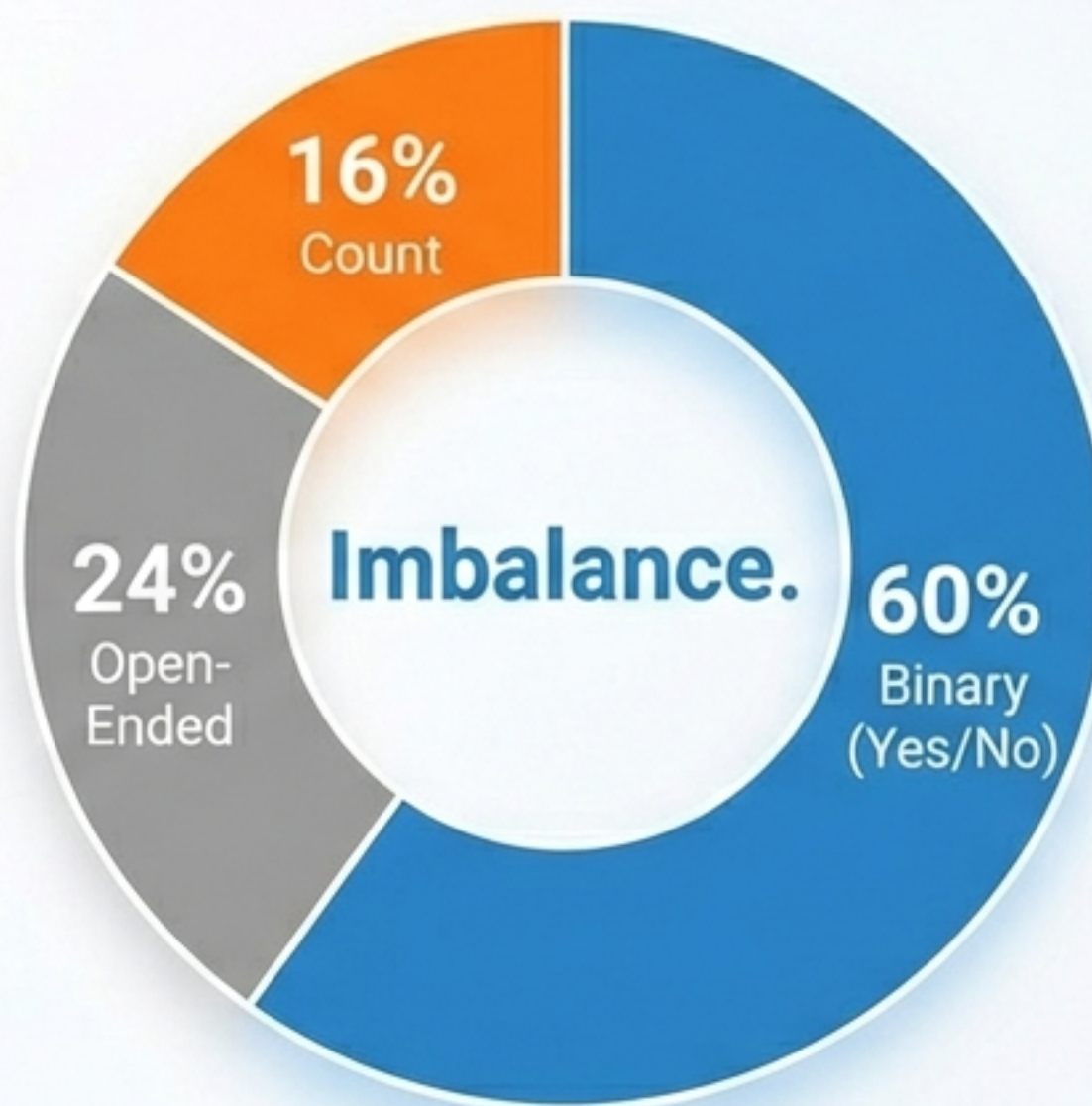
Total Images

3,515

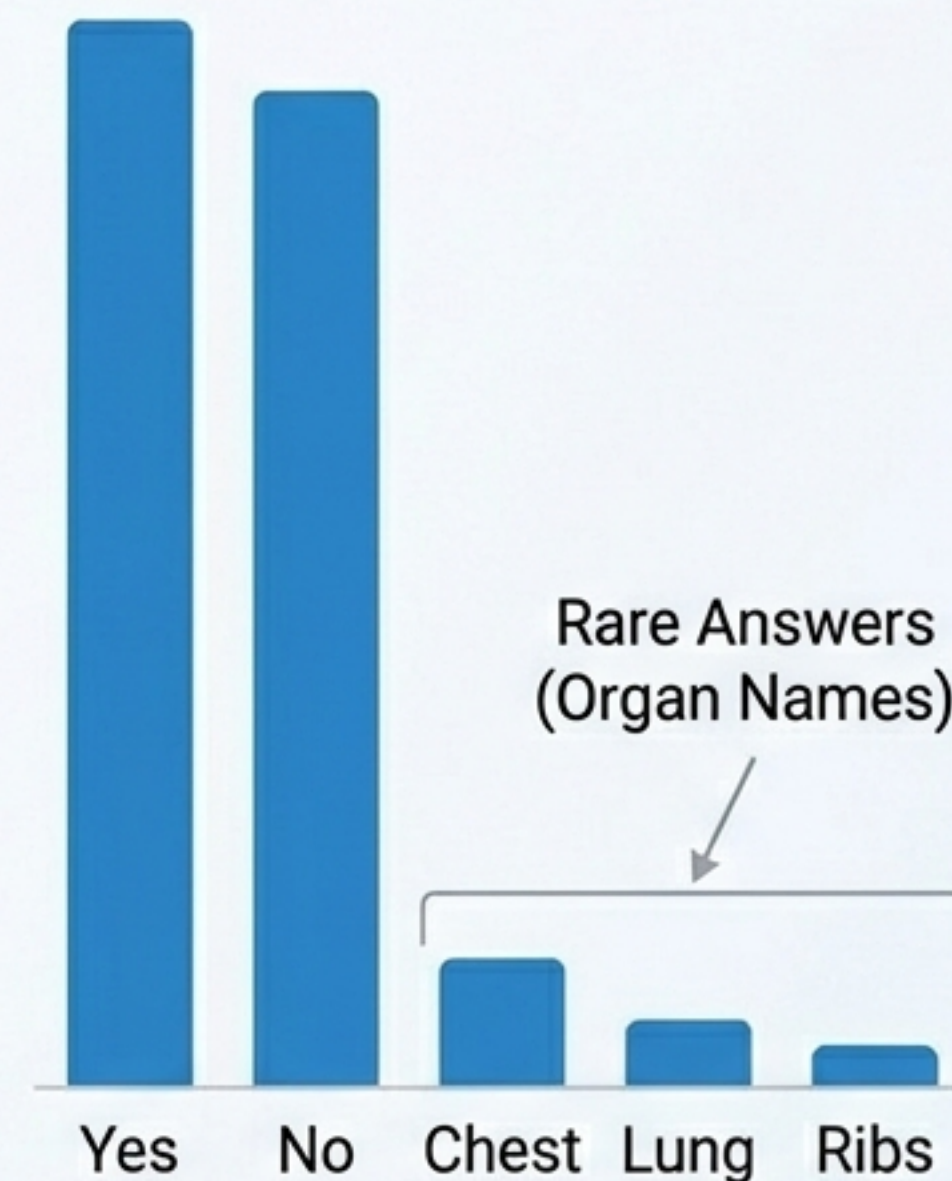
QA Pairs

Small-scale dataset requiring heavy augmentation.

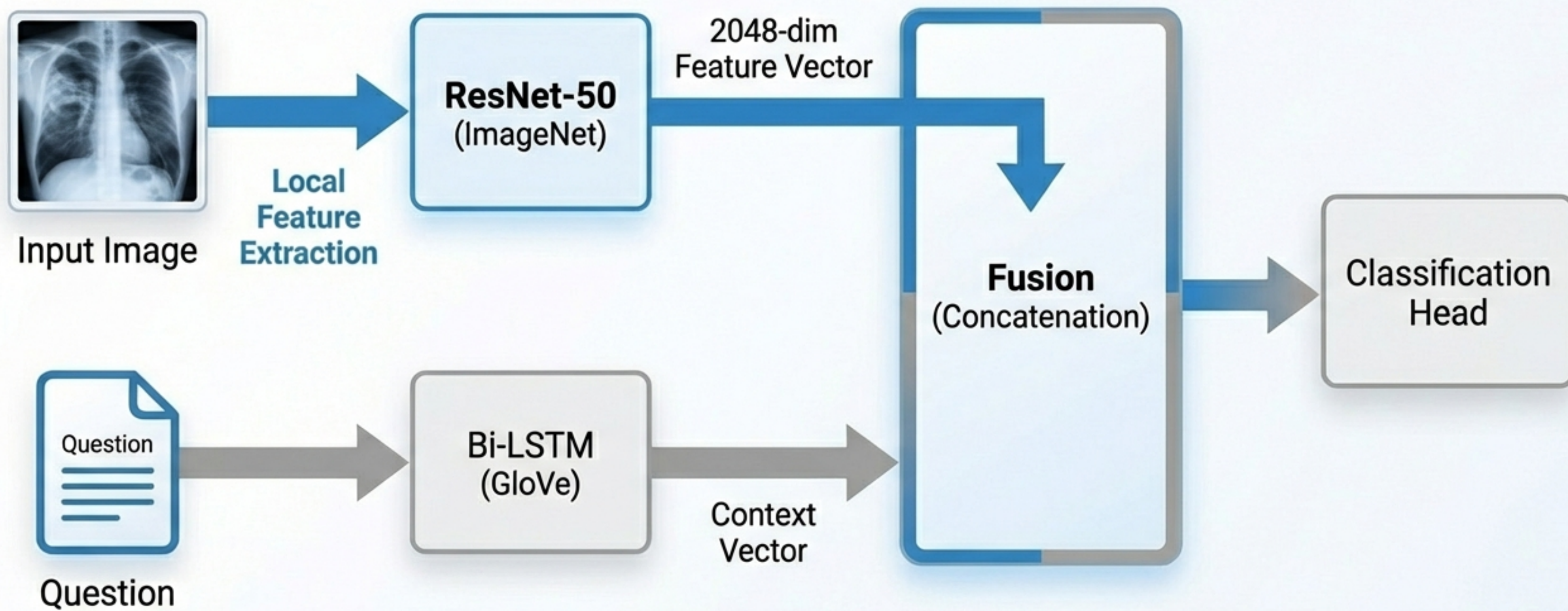
Question Type Distribution



The Long Tail

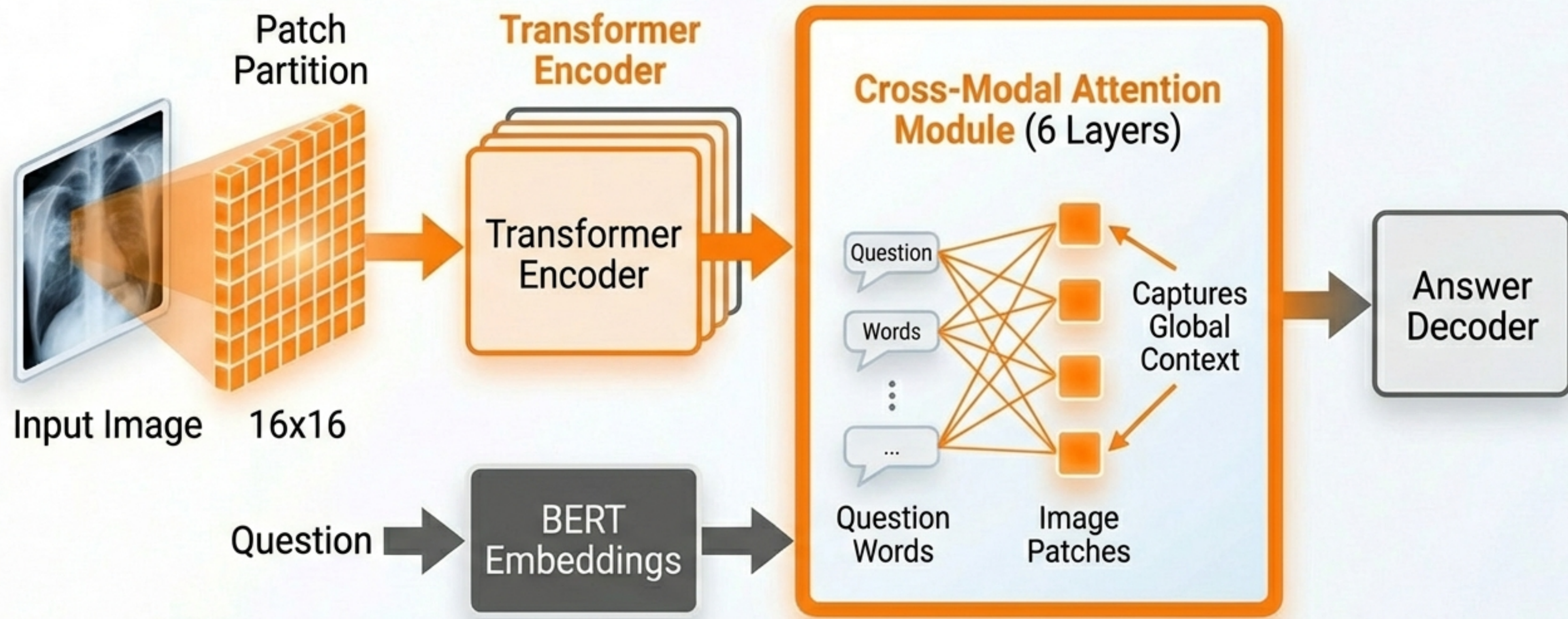


The Baseline: CNN-LSTM Architecture



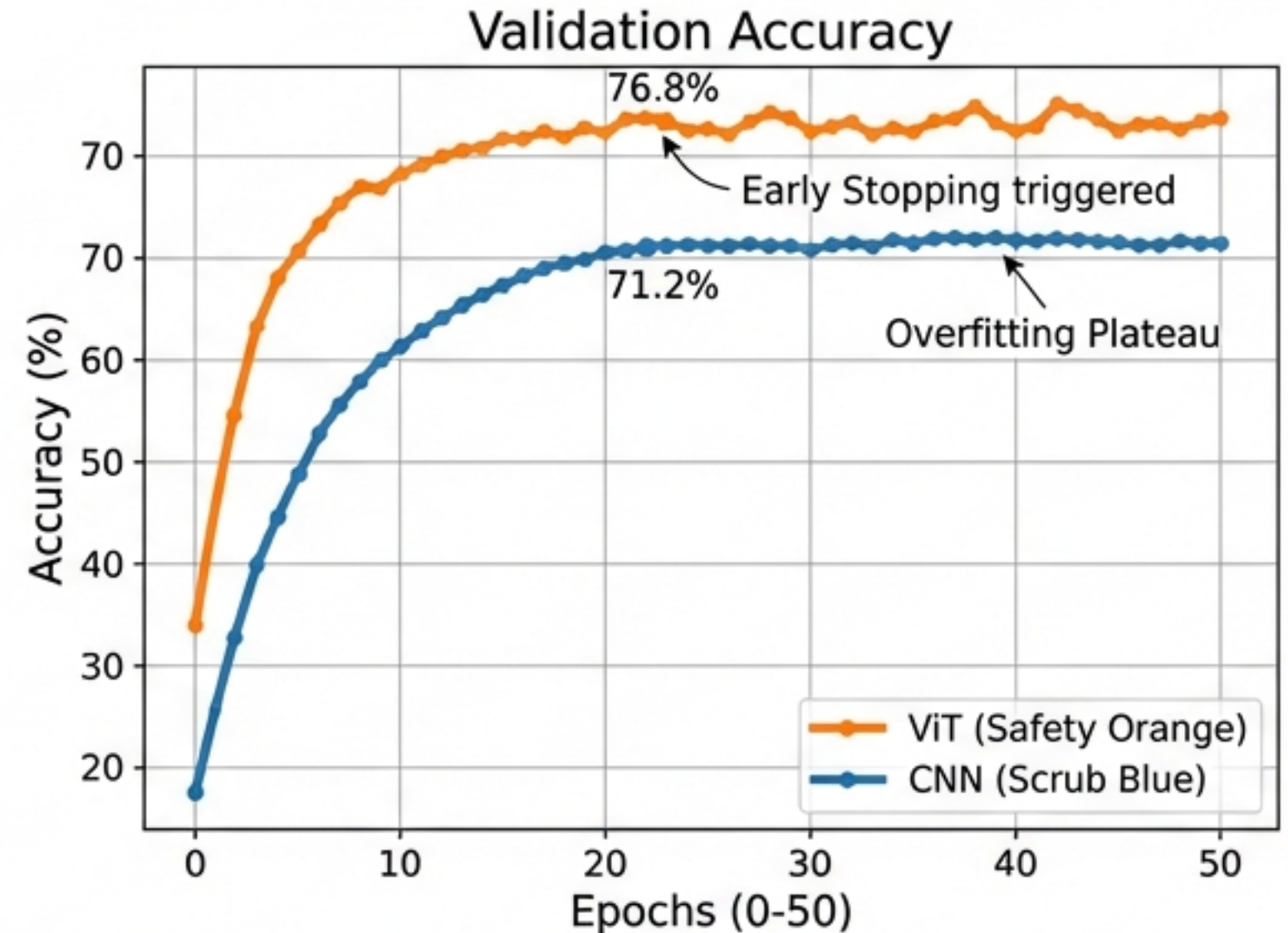
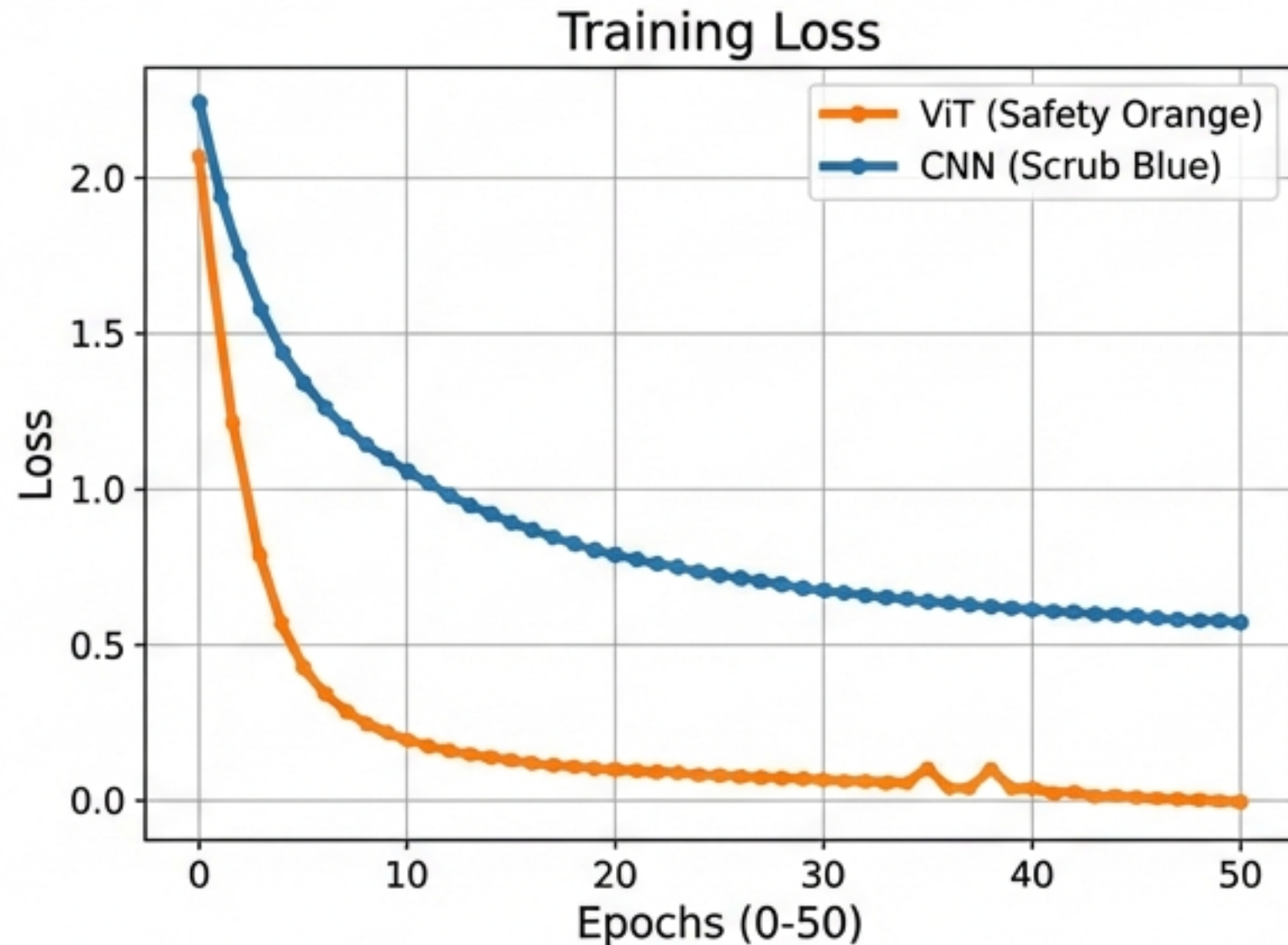
Total Parameters: 52.1 Million

The Proposed Solution: Vision Transformer (ViT)



Total Parameters: 86.4 Million

Training Dynamics & Convergence



ViT converges faster with better generalization using AdamW optimizer.

Head-to-Head Performance Results

71.2%

CNN Baseline

+5.6%
Improvement

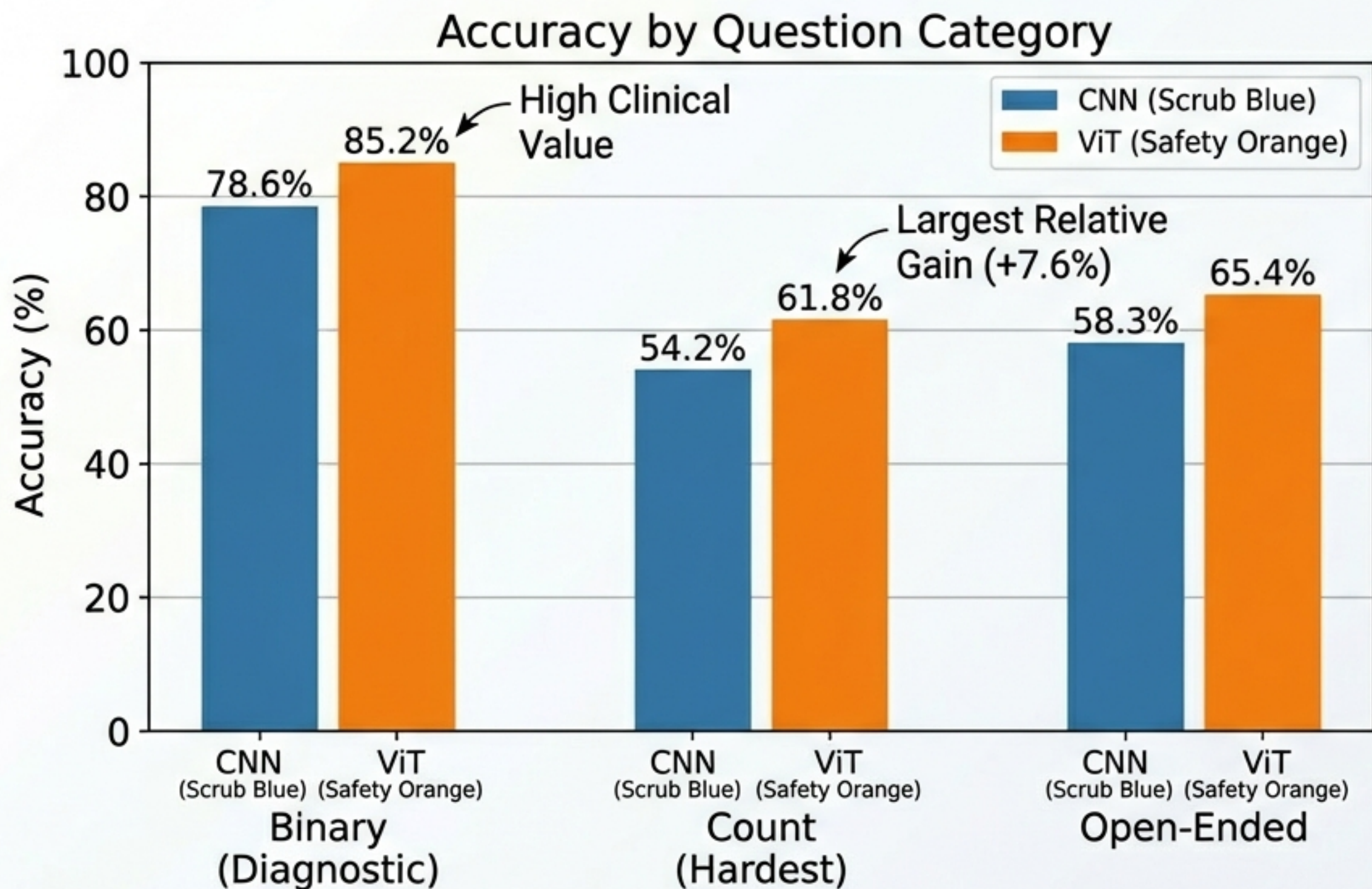
76.8%

Vision Transformer

Metric	CNN	ViT
Weighted F1-Score	0.71	0.77
Precision	0.69	0.75
Recall	0.67	0.73

Vision Transformer establishes new SOTA for VQA-RAD dataset.

Accuracy by Question Category



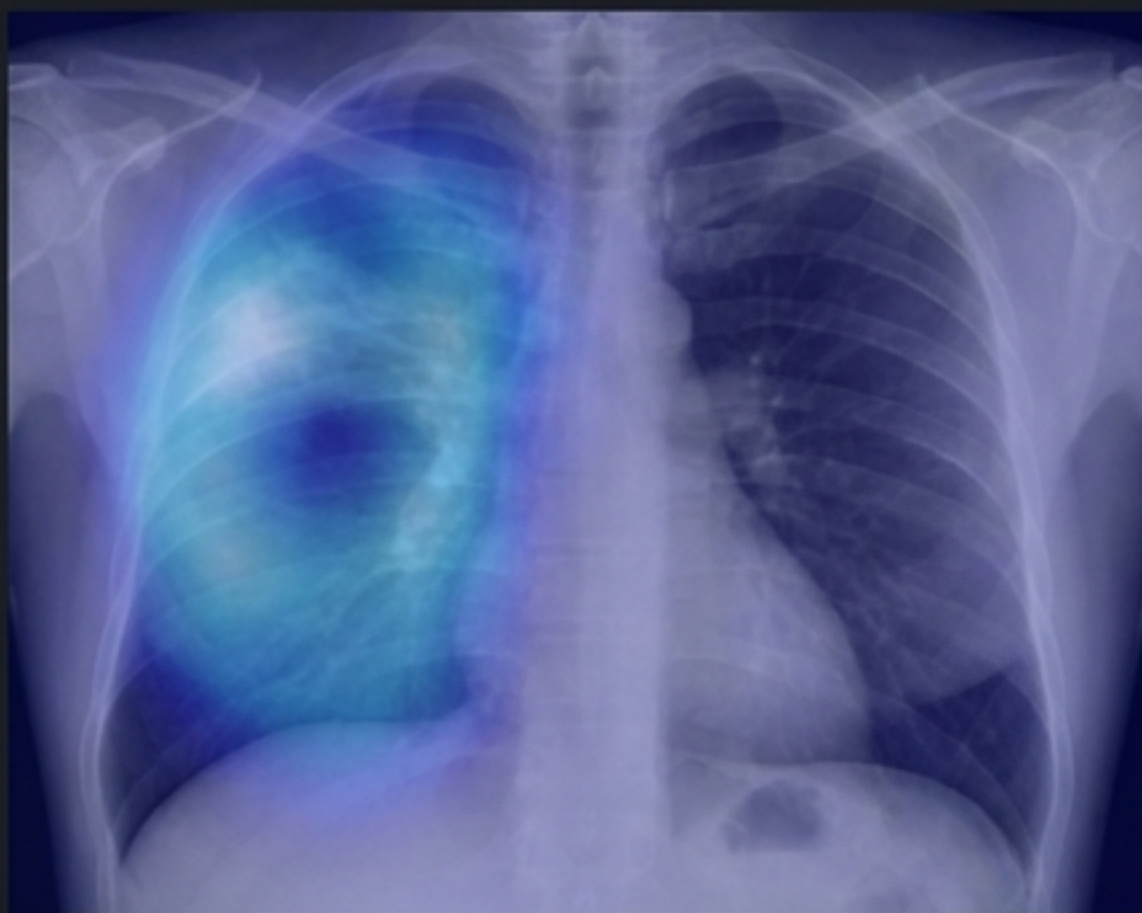
Interpretation:

- Binary accuracy (>85%) is sufficient for triage.
- Counting remains the primary bottleneck for both architectures.

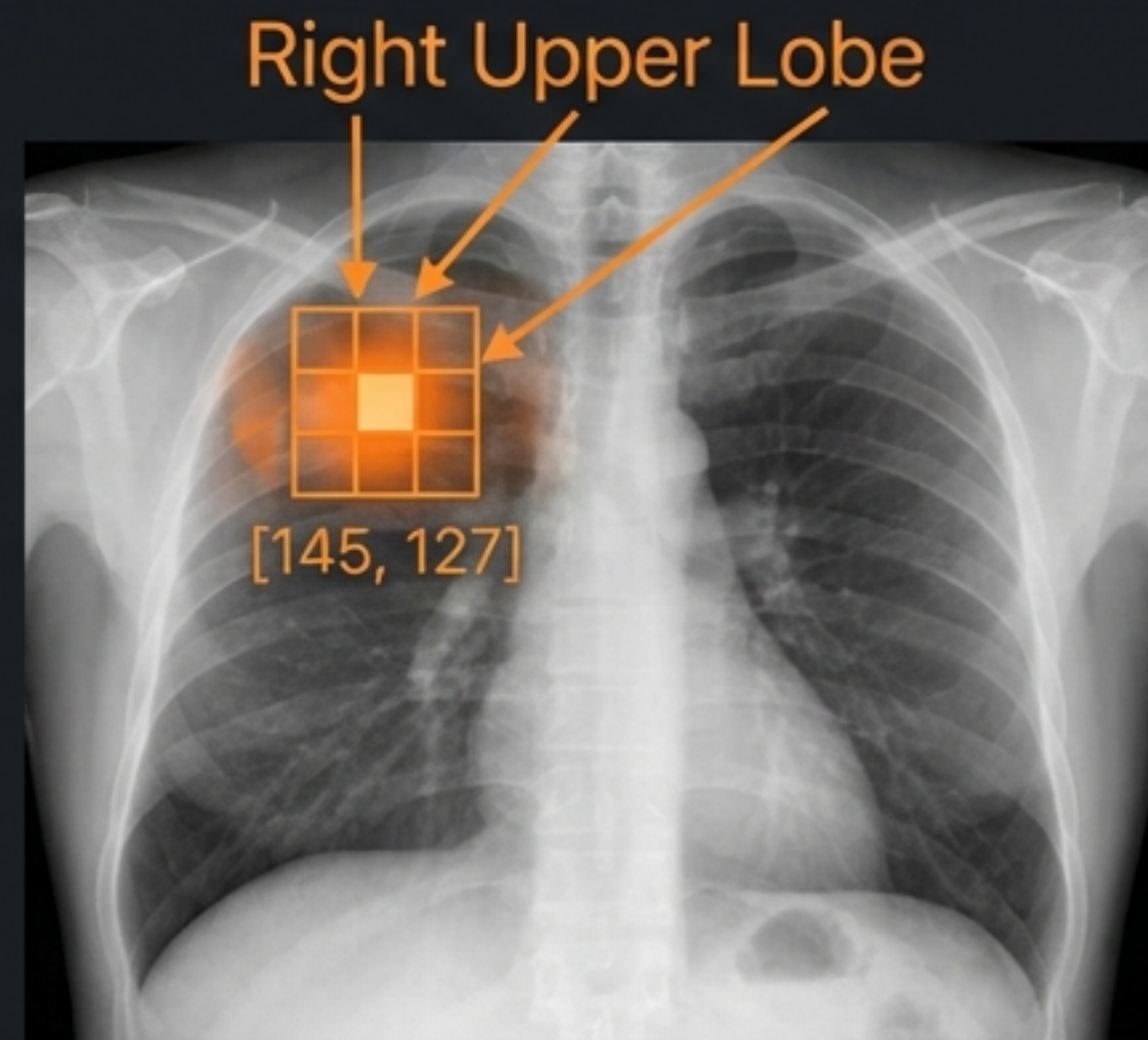
Interpretability: Seeing What the Model Sees

Question: "Is there pneumonia in the right upper lobe?"

CNN Attention



ViT Cross-Modal Attention



ViT explicitly aligns semantic concepts with visual pathology.

Error Analysis & Limitations

The Counting Problem

Confusion Matrix

	1	2	3	4
1	7	1	0	0
2	1	7	3	4
3	2	2	3	1
4	0	1	1	2
	1	2	3	4

Failure Modes

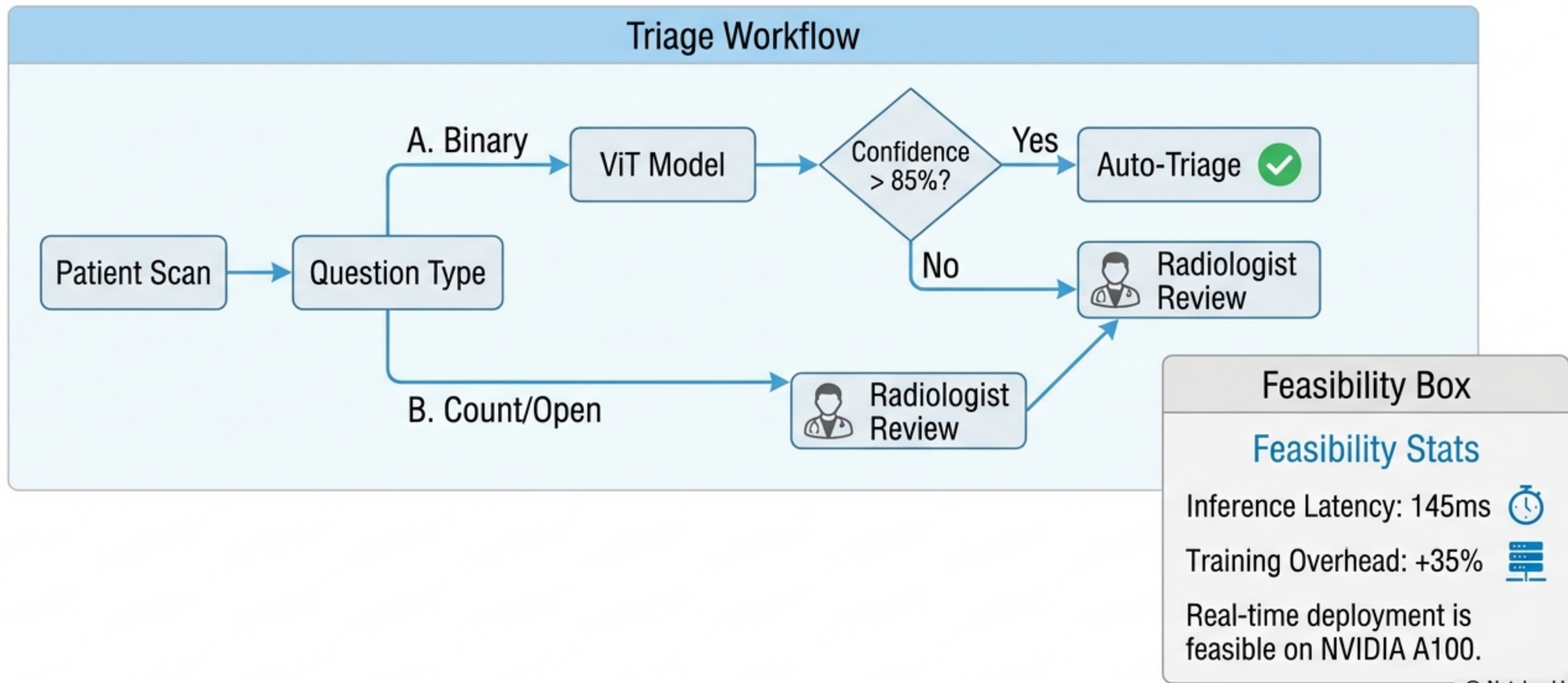


- **Complex Counting:**
Overlapping structures (ribs) confuse the model.
- **Fine-Grained Details:**
Lesions < 5mm are often missed.
- **Semantic Ambiguity:**
Synonyms are penalized.

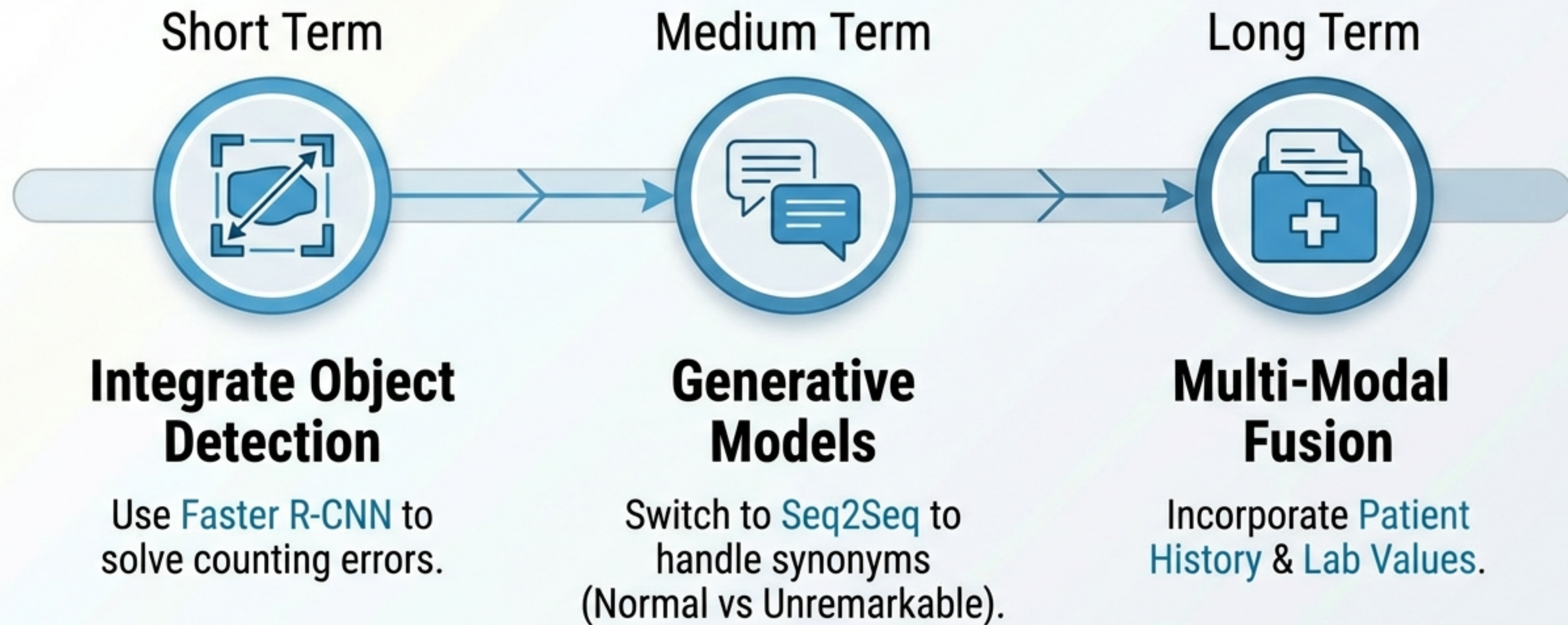
Normal
≠
No Abnormality

Both models lack explicit Object Detection (e.g., YOLO) capabilities.

Clinical Feasibility & Deployment



Future Directions



Team Contributions

Zhang Jiarui

- 💡 ViT Architecture Design
- 💡 Patch Embedding Implementation
- 💡 Data Preprocessing Pipeline

Ren Xingwang

- 💡 CNN Baseline Implementation
- 💡 Experimental Framework & Tuning
- 💡 Visualization & Error Analysis

Conclusion



The Victory

ViT outperforms CNN
Baseline (+5.6%).



The Mechanism

Global Context (Attention) is
superior to Local Features
for medical reasoning.



The Verdict

Vision Transformers
represent a new standard
for Medical VQA.

Moving from 'Black Box' classification to 'Trusted Assistant' reasoning.

Thank You.

We are happy to take questions.



github.com/adljh8181/WOA7015-AlternativeAssignment-MedVQA